Traffic Flow and Congestion Propagation Model for Huntsville, Alabama

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1. Introduction

Urban transportation systems are complex, interconnected networks that must balance high demand with limited infrastructure capacity. In rapidly growing cities such as Huntsville, Alabama, increasing commuter flows have led to recurring congestion, particularly along major corridors such as Interstate 565 (I-565). While congestion is often treated as a simple capacity issue, it is increasingly recognized as a systems-level phenomenon influenced by network structure, redundancy, and variability in demand.

This study develops a network-based simulation model to analyze traffic flow, congestion propagation, and system resilience in Huntsville. The objective is to evaluate how disruptions and demand variability affect system performance and to identify whether targeted infrastructure improvements can enhance network robustness.

2. Methodology

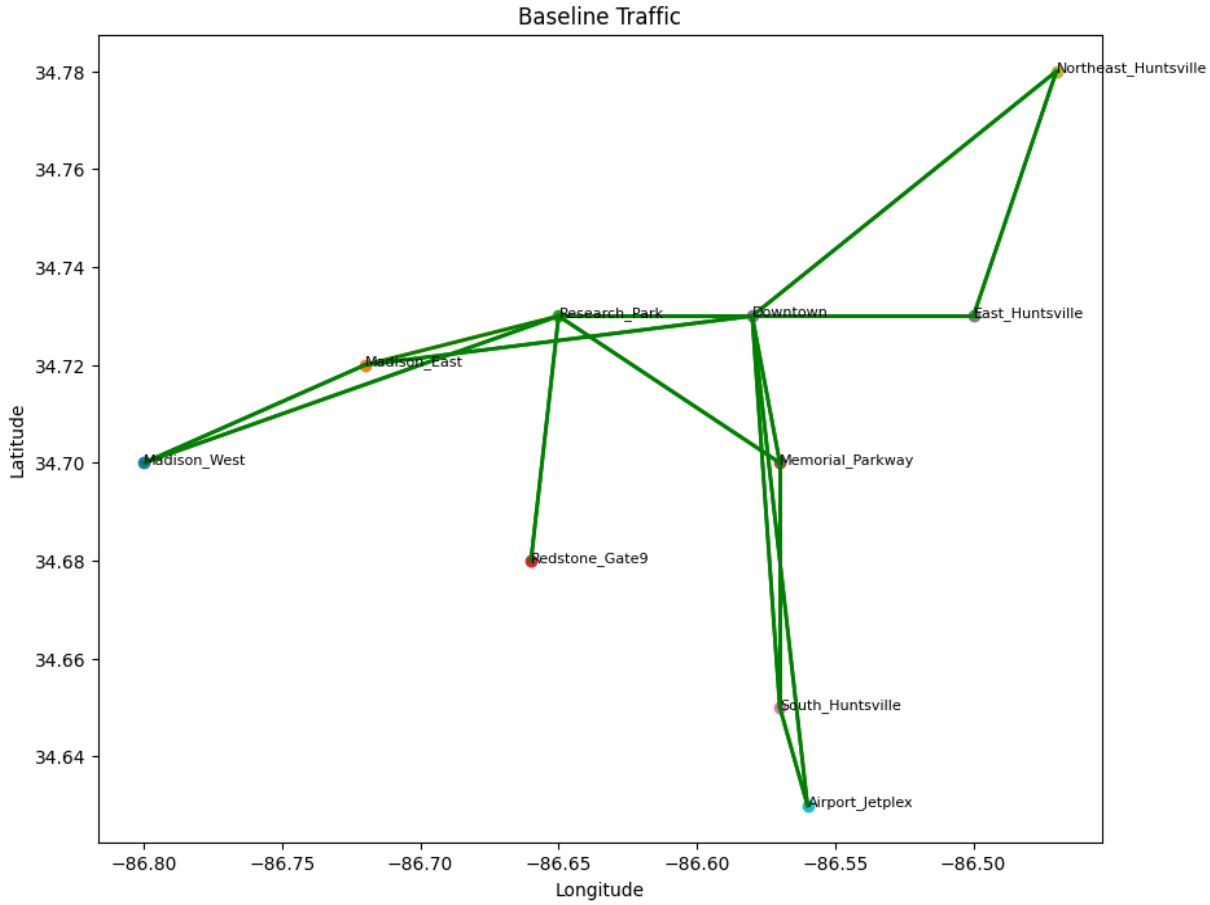
The Huntsville transportation system was modeled as a directed graph consisting of major zones (nodes) and primary corridors (edges). Each edge includes free-flow travel time, capacity, and length.

Origin–destination demand was synthetically generated to represent commuting patterns and adjusted for time-of-day variation.

Traffic assignment was performed iteratively using shortest-path routing, with congestion modeled using a volume-delay function. Monte Carlo simulation was used to introduce variability in demand and disruptions.

Time-of-day scenarios were also modeled, including AM Peak, Midday, PM Peak, and Night conditions, with different demand levels and incident probabilities. Figure 1 shows the baseline traffic map for Huntsville.

Figure 1. Baseline traffic in Huntsville, Alabama



3. Results

Baseline delay was approximately 124 vehicle-hours. Under a disruption scenario affecting I-565, delay increased to 2427 vehicle-hours (+1851%), demonstrating nonlinear system behavior. This result highlights the system's sensitivity to disruption and confirms the presence of critical capacity thresholds within the network.

A targeted improvement to an alternate corridor restored system delay to baseline levels, indicating the effectiveness of strategic redundancy. Figure 2 shows the Huntsville traffic map after an incident, while Figure 3 displays the map after the targeted improvement plan is implemented.

Figure 2. Huntsville traffic map after an incident

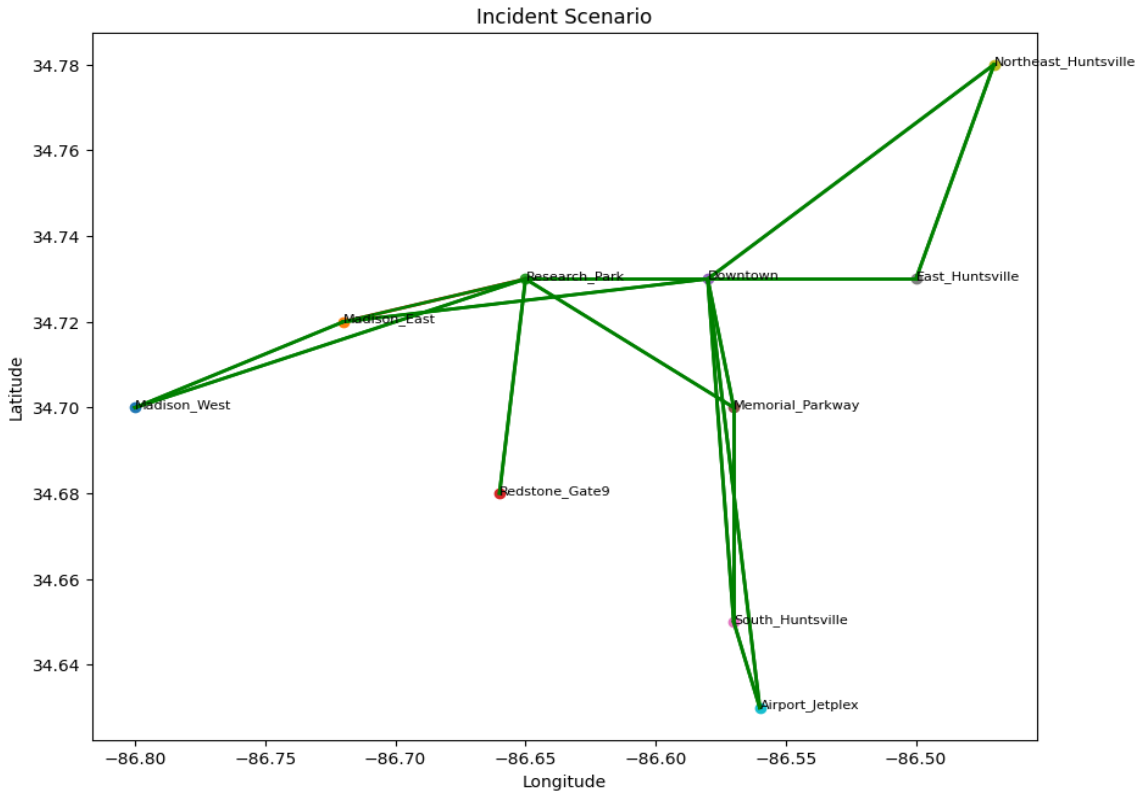
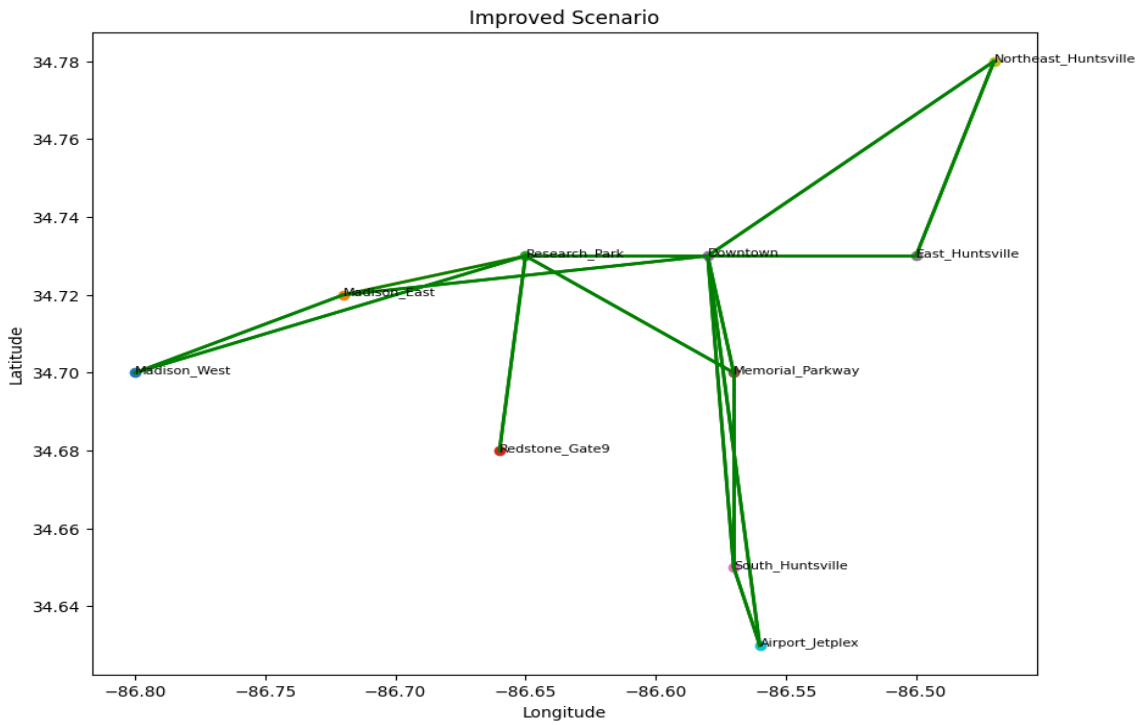
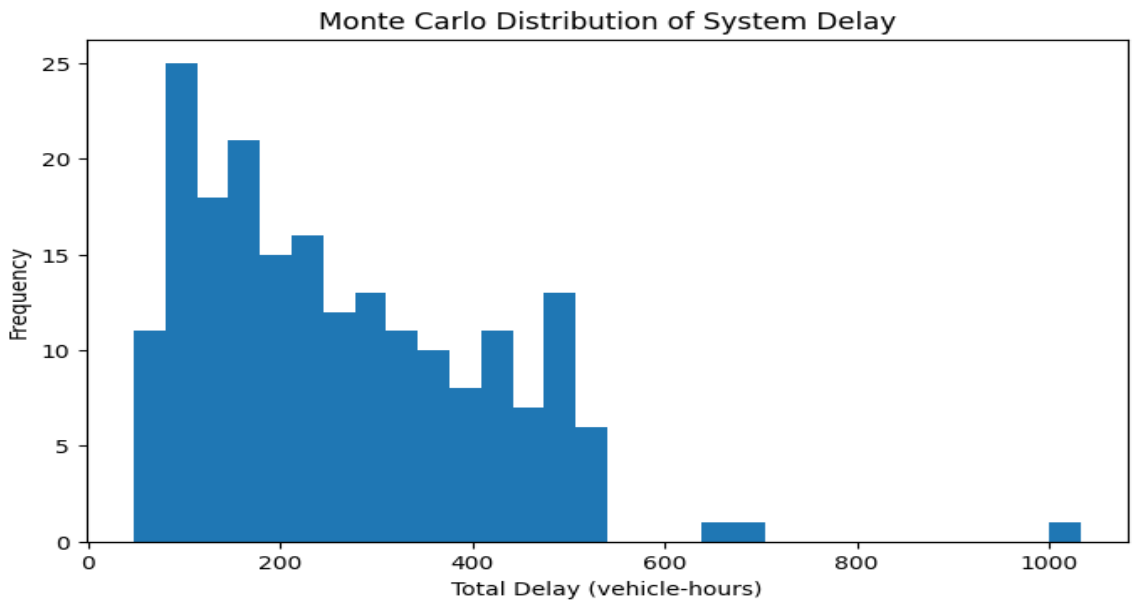


Figure 3. Huntsville traffic map with targeted improvement



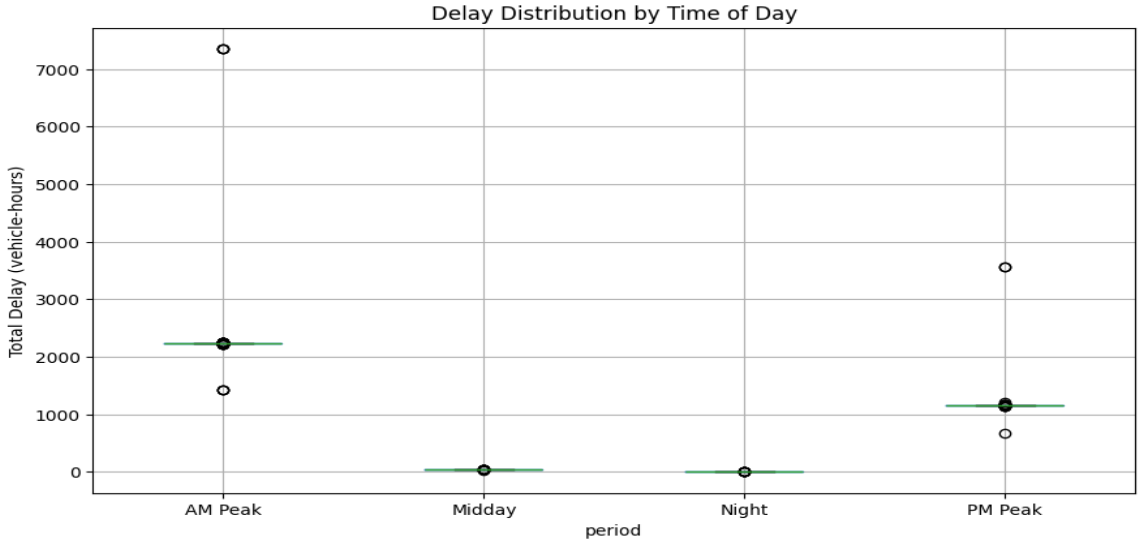
Monte Carlo simulation showed a right-skewed distribution of delay with a failure probability of approximately 6%, as Figure 4 below demonstrates. This indicates a low-frequency, high-impact failure system, where most conditions are manageable but those rare disruptions produce abnormally large congestion effects.

Figure 4. Distribution of system delays (200 simulations)



Time-of-day analysis showed that AM Peak produced the highest delays (~2411 vehicle-hours), followed by PM Peak (~1210), while Midday and Night remained stable, as shown by Figure 5 below. This behavior is likely due to the amount of traffic going to centralized employer hubs such as Redstone Arsenal, Research Park, and Downtown, where PM peak traffic is distributed among more disparate localities.

Figure 5. Distribution of delays in Huntsville traffic by time of day



3.1 Key Findings

- The Huntsville network is highly dependent on I-565, creating a critical vulnerability
- Congestion exhibits nonlinear behavior, with disruptions causing up to +1851% delay
- Targeted improvements to alternate corridors can fully restore system performance
- Peak congestion is time-dependent, with AM Peak representing the highest-risk period

4. Discussion

The network is highly dependent on I-565, making it vulnerable to disruptions. Congestion behavior is nonlinear, with small capacity changes leading to large impacts. This nonlinear behavior indicates that congestion does not increase proportionally with demand, but instead escalates rapidly once critical capacity thresholds are exceeded.

Targeted improvements to alternate routes significantly improve resilience, suggesting that strategic redundancy is more effective than broad expansion.

Time-of-day results indicate that inbound commuter flows during the AM peak create greater system stress than outbound flows in the PM peak.

5. Conclusion

Huntsville's transportation network operates near capacity and is highly sensitive to disruptions. While severe congestion events occur infrequently, their impact is substantial, indicating a system that is stable under normal conditions but highly vulnerable to disruption.

Strategic improvements to key alternate corridors can dramatically improve resilience and mitigate system-wide congestion.

This study highlights the importance of network structure in transportation planning and provides a framework for evaluating infrastructure investments.

6. Future Work

Future work includes incorporating real traffic data, adding adaptive routing behavior, expanding to multi-modal systems, and applying machine learning to predict congestion risk.